

Multilevel Inheritance and Method Overloading Program

Program Code:

```
class A {
    void smtg() {
        System.out.println("This is Class A");
    }

    void add() {
        int a = 10, b = 20;
        System.out.println("Addition is: " + (a + b));
    }
}

class B extends A {
    void smtg2() {
        System.out.println("This is Class B");
    }

    void add(int c) {
        int a = 10, b = 20;
        System.out.println("Addition is: " + (a + b + c));
    }
}

class C extends B {
    void smtg3() {
        System.out.println("This is Class C");
    }

    void add(int c, int d) {
        int a = 10, b = 20;
        System.out.println("Addition is: " + (a + b + c + d));
    }
}

public class Main {
    public static void main(String[] args) {

        C obj = new C();

        obj.smtg();
        obj.smtg2();
        obj.smtg3();

        obj.add();
        obj.add(30);
        obj.add(30, 40);
    }
}
```

Explanation:

This program demonstrates multilevel inheritance and method overloading. Class B extends class A, and class C extends class B. Therefore, an object of class C can access the methods of classes A, B, and C. The add() method is overloaded with zero, one, and two parameters. Each overloaded method performs addition using the given values.

Compilation and Execution Commands:

Save the file as: Main.java

Compile:
javac Main.java

Run:
java Main

Output Screenshot:

```
This is Class A
```

```
This is Class B
```

```
This is Class C
```

```
Addition is: 30
```

```
Addition is: 60
```

```
Addition is: 100
```

Verification:

The output confirms that the object of class C successfully accesses methods from class A, class B, and class C. The three overloaded add() methods produce the correct results: 30, 60, and 100.

Result:

Thus, the Java program was successfully executed. Multilevel inheritance and method overloading were verified successfully.